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Subject 1- Cyber Security

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Assignment

Question Paper

Cyber Security

Q.1. Attempt any Eight of the Following.

a) What is Cyber security ?

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Cyber Security refers to the practice of protecting systems, networks, and programs from digital attacks aimed at accessing, changing, or destroying sensitive information, extorting money, or disrupting normal business processes.

Cyber Security is the set of principles and practices designed to protect our computing resources and online information against threats.

b) What is Virus ?

→

A virus is a type of malicious software that, when executed, replicates by inserting its code into other computer programs, data files, or the boot sector of the hard drive. When this replication is successful, the affected areas are said to be 'infected'.

c) What is attack vector ?

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An "attack vector" is a path, which an attacker can gain access to a computer or a network server to deliver a malicious outcome. Attack vectors enable attackers to exploit system vulnerabilities, including the human element.

- 1) Attack by E-mail.
- 2) Attachments
- 3) Attack by deception
- 4) Hackers
- 5) Headless guest
- 6) Attack of the worms.
- 7) Malicious macros
- 8) Foistware (Sneakware)
- 9) Virus

c) State Social media marketing.

Social media marketing is the use of social media platform to connect with an audience to build a brand, increase sales, and drive website traffic. This involves publishing content on social media profiles, engaging with followers and analyzing result.

c) What is Steganography?

Steganography is the practice of hiding a secret message inside of something that is not secret. The hidden data can be text, image, audio, or even video content.

Steganography Types.

- 1) Text Steganography
- 2) Image Steganography
- 3) Video Steganography
- 4) Audio Steganography
- 5) Network Steganography.

Differentiate between virus and Worm.

Virus	Worm
1) A Virus requires a host program to spread and usually corrupts or modifies files when it infects them.	1) A Worm is standalone malware that replicates itself and spreads across networks without needing to attach to a host program or require user intervention.
2) Require human action. Such as running an infected file or program to spread.	2) Automatically replicates itself and spreads over networks or the network.
3) Needs to attach itself to a specific file or program to function.	3) Does not need to attach to a file or program and functions independently.
4) Example - Melissa virus	4) Example :- Morris worm, Dard

g) Define Footprinting.

Footprinting is the process of gathering as much information as possible about a target system, network, or organization in a legal manner, often as a preliminary step before a cyber attack.

Footprinting Types:

1) Active Footprinting.

2) Passive Footprinting.

h) What is Cyber Stalking?

Cyber stalking is the use of the internet or other electronic means to stalk or harass an individual, group, or organization. It may include monitoring, identity theft, threats or gathering personal information without consent.

Types of Stalkers

1) Online stalkers.

2) Offline stalkers.

i) What is phishing?

"Phishing" refers to an attack using mail programs to cheat internet users and disclosing confidential

Information that can be then utilized for illegal purpose. phishing is a type of Cyber Security threat. It targets users directly through email text or direct Message.

Phishing Techniques

- 1) Email Phishing.
- 2) Spear Phishing.
- 3) Smishing.
- 4) Whaling.
- 5) Vishing.

J. Define the term Cyber Security

Cyber Security refers to the practice of defending Computer Servers, mobile devices, electronic System, networks, and data from malicious attacks. It includes a range of technologies, processes, and practices designed to protect networks, devices, programs, and data from attacks, damage, or Unauthorized access.

K. What is Intellectual Property.

Intellectual Property (IP) refers to Creations of the mind such as Inventions, literary and artistic works, designs, Symbols, names, and images used in Commerce. IP is protected by law through patents, copyrights, trademarks and trade secrets.

Q.2. Attempt any four of the following.

a) Differentiate between Active attack and Passive attack.

Active Attack	Passive Attack
1) An active attack involves actively attempting to alter or disrupt communication, data, or system.	1) A passive attack focuses on monitoring or eavesdropping on communication without altering the system or data.
2) The attacker seeks to modify, delete, or inject malicious data into the system.	2) The attacker seeks to gain unauthorized information without the knowledge of the system owner.
3) Easier to detect because the system's behavior is changed.	3) Difficult to detect because no data is altered.
4) Example - Man-in-the-Middle, Data Modification	4) Example - Eavesdropping, Traffic Analysis.

b) Explain the Cyber security real life incident example.

→ Equifax Data Breach (2020): In one of the most significant Cyber Security breaches, the credit reporting agency Equifax was attacked, leading to the exposure of personal

data of over 147 million people. Hackers exploited a vulnerability in Equifax's web application software, gaining access to names, Social Security numbers, birthdates, and addresses. The breach highlighted the need for regular software updates and patches to prevent attackers from exploiting known vulnerabilities.

Q) Discuss IPR Issues.

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IPR stands for Intellectual Property Rights.

1) Software Piracy :- Illegally copying or distributing software can lead to violations of intellectual property laws, harming developers and businesses financially.

2) Data Theft :- Stealing proprietary or sensitive business information, such as algorithm research data, or trade secrets, is a critical IPR issue.

3) Cyber Espionage :- Hackers targeting businesses to steal patents, designs or other intellectual property for competitive advantage can cause significant damage.

4) Copyright Infringement :- Unauthorized use of digital content (music, video, software) without the owner's consent, facilitated by the Internet, is a widespread problem. Protecting copyrighted materials from online theft is a key cyber security concern.

d) What is SQL injection and what are the different Countermeasures to prevent the attack?

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SQL, or Structured Query Language, is the standard language for interacting with relational database. In apps and other types of programming, database are used to store user data such as usernames and passwords. Database are also often the most effective, secure solution for storing other types of data from public blog posts and comments to confidential bank account numbers.

SQL Injection (SQLi) allows attackers to add, edit, and delete notes from the database. SQL Injection is a security defect on a database that can impact web applications and websites that use SQL database like SQL Server, MySQL, and Oracle.

Prevent SQL Injection:

1) Use of Prepared Statements / Queries :- Parameterized queries are a means of pre-compiling an SQL statement.

2) Escape all user supplied input :- Always use character-escaping functions for user supplied input provided by each database management system.

3) Use of Stored Procedures in database :- Stored procedures require the developer to group one or more SQL statements into a logical unit to create an execution plan.

4) Apply Least Privilege :- It's better to enforce least privilege

On the database to defend the application against SQL Injection.

5) Isolate database Server with Web Server:- This provides added layers of protection that shield us from the negative effects of bad code, human error, or a security vulnerability.

e) Why do we need Cyber laws : The Indian Context ?
→

Cyber law give legal recognition to all risks arising out of the usage of computers and its network. Cyber law covers various aspects like intellectual property, data protection and privacy, freedom of expression and crimes committed using computers. The Indian Parliament passed its first cyberlaw, the ITA 2000. The main purpose is to provide the legal infrastructure for e-commerce in India.

The need reason for enactment of cyberlaw in India are:

India lacks in many aspects when it comes to newly developed Internet technology. There is a need to have some legal recognition to the internet as it is one of the most dominating sources of carrying out business in today's world. With growth of Internet, a new concept called Cyber Terrorism came in picture. This Cyberterrorism activity affects on the social, ideological, religious, political or cyber similar objectives. Keeping all these factors into consideration, Indian parliament passed the Information Techⁿ Bill on 17 May 2000, known as the ITA 2000.

Q.3 Attempt any four of the following.

a) Discuss how emails are used in Forensics analysis.

→ Softwares are available that helps to Create Fake emails. The forensic analysis of Email is an important aspect of Cyber forensic analysis which helps to establish the authenticity of an Email when it is suspected. Forensic analysis of email, you must know the email Component and the email header structure.

An Email System is the hardware and Software that Controls Flow of E-mail. Two main Components of an E-mail System are Email Server and E-mail gateway.

E-mail Servers are Computer that forwards, collect, store and deliver email to their Clients.

1) Header analysis :- Investigating the metadata of email -s to track the Sender's IP address, timestamp, and route the email took through various Servers.

2) Content analysis :- Examining the body of the email, including any hidden messages, embedded links, or malware in attachments.

3) Attachment analysis :- Analyzing the files attached to emails for malicious Code or sensitive data.

4) Email recovery :- Retrieving deleted or archived emails from email Servers or local machines.

b) Explain different types of Credit Card Frauds.

Credit Card Fraud refers to the Unauthorized use of Someone's Credit Card Information for Financial gain.

1) Card-not-present (CNP) Fraud: This occurs during Online transaction where the physical card isn't required. Fraudsters use stolen card details to make purchases Online.

2) Skimming: A device is used to steal card information during physical transactions, often at ATMs or point of sale (POS) machines.

3) Phishing: Scammers trick individuals into revealing their card information through fraudulent emails, website or phone calls.

4) Account takeover: Hackers gain access to a cardholder's account by stealing their personal information, allowing them to make unauthorized transactions.

5) Lost or Stolen Cards: Physical theft of a credit card or using a lost card to make purchases without the owner's consent.

c) Explain the rules of Digital Evidence.

Digital evidence refers to any information stored or transmitted in digital form that can be used in court. The handling

ing of digital evidence is governed by certain rules to ensure its integrity and admissibility.

1) **Admissibility** :- The evidence must be relevant, legally obtained and presented in a way that is acceptable in Court.

2) **Integrity** :- The Original evidence must not be altered. Proper documentation and Chain of Custody must be maintained to ensure that the evidence remains untampered.

3) **Authenticity** :- It must be proven that the evidence is genuine and has not been modified since its collection.

4) **Reliability** :- The evidence should be collected using reliable methods and tools, and it must be clear that the evidence is accurate and trustworthy.

5) **Chain of Custody** :- The movement and handling of evidence should be documented thoroughly to establish its authenticity and prevent it from being questioned.

Q) What is domain Name ? Explain with example.

A domain name is the unique address used to identify a website on the internet. It is a human-readable form of an IP address, which directs users to the server where the website's content is stored. Domain names are part of the Domain Name System (DNS), which translates

domain names into IP addresses.

Ex. In the domain `www.example.com`, 'example.com' is the domain name, where '.com' represents the top-level domain (TLD) and 'example' is the second-level domain (SLD).

9) Prepare a Case study with its implication on "Company Website Hacked".

→ Case Study: Classmate Corp Website Hacked. Classmate Corp, a mid-sized e-commerce company, experienced a cyber attack where their website was compromised. Hackers gained unauthorized access to the company's web server, defaced the website, and stole customer data, including credit card information.

Investigation: A forensic investigation revealed that the hackers exploited a vulnerability in the website's content management system (CMS), which had not been updated with the latest security patches. Additionally, weak passwords were used for administrative accounts, making it easier for the attackers to gain access.

Implications:

1) Financial Losses: Classmate Corp faced significant financial losses due to a drop in customer trust and legal liabilities related to the breach of sensitive customer information.

2) Reputation Damage:- The Company's reputation was severely damaged, resulting in lost customers and reduced business opportunities.

3) Legal Consequences:- The Company faced lawsuits from affected customers and penalties from regulatory bodies for failing to protect personal data.

4) Operational Disruption:- The website had to be taken offline for repairs, resulting in a loss of sales and operational downtime.

Preventive Measures:- Classmate Corp implemented stronger cybersecurity protocols, including regular updates to software, stronger password policies, and routine security audits, to prevent future attacks.

Q.4. Attempt any four of the following.

a) → Explain Organizational guidelines for internet Usage.

The organization recognizes that there is a need for protecting the company's identity when online. Every time an employee accesses the internet, an employee may leak data to watchful company competitors. Also, hackers and online predators may leak company data. In such cases, the organization needs an effective solution. The organization must develop safety computing guidelines, which may be referred to as organization guidelines for the internet usage or computer usage policy.

1) Acceptable Use Policies (AUPs) : Define what employees can and cannot do with the company's Internet resources (e.g. downloading, software, visiting certain websites).

2) Monitoring and privacy : Employees should be aware that their internet usage may be monitored for security or performance purpose.

3) Security Guidelines : Prohibiting the use of unsecured network, sharing passwords, or downloading malicious software.

4) Personal Use : Policies typically allow limited personal use of the internet but prohibit activities that interfere with job performance or violate legal or ethical standards.

b) What are the challenges to Indian law and Cybercrime Scenario in India?

In that context we found that the Indian IT law does not provide clear definition to the term Cybercrime. The Indian Penal Code (IPC) does not use the Cybercrime term even after its amendment by the ITA 2008. On the contrary, it has a separate.

1) Jurisdiction Issue : Cybercrimes often transcend national borders, making it difficult to determine which country's law apply and how to enforce them.

2) Lack of awareness : Many individual and businesses

are not fully aware of Cyber laws, leading to insufficient protection and Compliance.

3) Evolving Threats :- As technology evolves new types of Cybercrimes emerge. Such as ransomware, phishing, and identity theft, requiring constant updates to laws and enforcement mechanisms.

4) Law Enforcement Capabilities :- There is a shortage of Skilled professionals in law enforcement agencies who can effectively investigate and Prosecute Cybercrimes.

C1 → Discuss Various password Cracking techniques

Passwords are like a key to get an entry into Computerized Systems. password cracking is a process of recovering password from data that has been stored in or transmitted by a Computer system in a scrambled form.

1) Dictionary Attack :- A dictionary file is loaded into the Cracking application that runs against user accounts. Attacker tries all dictionary words before trying Brute force attack.

2) Brute Forcing Attack :- The program tries every combination of character until the password is broken.

3) Rule-based Attack :- This Attack is used when the attacker gets some information about the password.

4) Password guessing - The attacker creates a list of all possible passwords from the information collected through social engineering or any other way and tries them manually on the victim's machine to crack the password.

5) Hash Injection and phishing - Attacker injects a compromised hash into a local session and uses the hash to validate the network resources.

6) Trojan / spyware / keyloggers - Attacker installs Trojan / spyware / keyloggers on victim's machine to collect victim's usernames and passwords. Then it runs in the background and sends back all user credentials to the attacker.

7) Passive Online Attacks - Attacker performs password cracking without communicating with the authorizing party.

Examples :-

- 1) Wire sniffing
- 2) Man-in-the-Middle.
- 3) Replay.

8) Offline Attack : Attacker copies the target's password file then tries to crack password in his own system at different locations.

Example :-

- 1) Pre-computed Hashes (Rainbow Table)
- 2) Distributed Network.

- 4) Non-Electronic Attacks: Attackers need not possess technical knowledge to crack password, hence known as Non-technical attacks.

Examples:-

- 1) Shoulder Surfing:-
- 2) Social Engineering
- 3) Dumpster Diving.

- d) Explain CIA triad.

→

Confidentiality, Integrity and Availability, also known as the CIA triad, is a model designed to guide policies for Information Security within an organization. Confidentiality, integrity and availability together are considered the three most important concepts within info. security. These three principles together can help guide the development of security policies for organization.

1) Confidentiality: Confidentiality measures are designed to prevent sensitive information from unauthorized access attempts. Here, data is categorized according to the amount and type of damage that could be done.

2) Integrity: Maintaining the consistency, accuracy and trustworthiness of data over its entire lifecycle. Data must not be changed in transit, and steps must be taken to ensure data cannot be altered by unauthorized people.

3) Availability :- Availability means Information should be consistently and readily accessible for authorized parties. This involves properly maintaining hardware and technical infrastructure and system that hold and display the information.

Q → Explain various types of Cyber Forensics.

Cyber Forensics, also known as digital forensics, is the practice of collecting, analyzing and preventing electronic data in a way that is admissible component of Cyber security, helping to investigate and resolve cybercrimes.

1) Computer Forensics :- Involves identifying, preserving, analyzing, and reporting data from computer and other digital devices.

Usages :- It is primarily used to investigate computer system suspected of being involved in cyber crimes like hacking, fraud, or theft of data.

2) Network Forensics :- Focuses on monitoring and analyzing computer network traffic to detect and respond to intrusions or malicious activities.

Usages :- Commonly used in cases of network breaches where the data packets, logs, and network events are analyzed to trace the origin of attacks.

3) Mobile Device Forensics! - The process of recovering and analyzing data from mobile devices, such as smartphones and tablets.

Usage! - It involves extracting call logs, text messages, photos, videos.

Q.5. Write a short note on any two of the following.

1) → The Indian IT Act.

The Indian IT Act was published in the year 2008 with the purpose of providing legal recognition for processes carried out by means of electronic data exchange and other means of electronic communication, generally referred to as E-Commerce.

Electronic Communications involve the use of alternative to paper-based methods of communication and storage of data, to facilitate electronic filing of documents with the government agencies. Another intention of the Indian IT Act was to amend the Indian Penal Code (IPC), the Indian Evidence Act 1872, the Bankers' Books Evidence Act 1891, the Reserve Bank of India Act 1934.

b) Need of Cyber Laws.

Cyber Law give legal recognition to all risks arising out of the Usage of Computers and its Networks.

Cyberlaw Covers Various aspects like intellectual property, data protection and privacy, freedom of expression and Crimes Committed Using Computers.

The Indian Parliament passed its first Cyberlaw, the ITA 2008. The main purpose is to provide the legal infrastructure for E-commerce in India.

c) Social Media Marketing.

The social computing and social media marketing are same. Importance of Social media is faster growing. According to the 2020 survey by Marketing professionals Usage of Social media site by large Business to Business B2B Organization show.

